

LONG POINT, ON, Canada

WMO: 714640

Lat: 42.549N

Lon: 80.049W

Elev: 175

StdP: 99.24

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-14.9	-12.3	-20.8	0.6	-15.5	-18.7	0.7	-13.7	20.9	-3.1	18.8	-3.2	9.0	230	0.735

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.7	26.2	21.5	25.3	21.1	24.6	20.7	22.7	24.9	22.1	24.3	21.5	23.8	5.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.0	17.0	24.5	21.2	16.2	23.9	20.5	15.5	23.4	68.2	24.9	65.6	24.5	63.3	23.8	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	16.0	14.2		-16.0	28.8	3.9	1.4	-18.8	29.7	-21.0	30.5	-23.2	31.3	-26.0	32.3
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	-2.7	-2.7	0.8	6.3	12.5	18.3	21.7	21.8	18.7	12.5	6.7	1.0
	DBStd	9.74	5.37	5.41	4.81	3.99	3.87	3.10	2.22	2.09	3.27	4.08	4.43	4.65
	HDD10.0	1596	393	357	287	125	17	0	0	0	0	22	115	281
	HDD18.3	3481	651	590	543	362	186	38	3	2	34	185	348	538
	CDD10.0	1465	1	0	2	13	96	248	363	364	261	99	17	1
	CDD18.3	306	0	0	0	0	5	36	107	108	46	4	0	0
	CDH23.3	553	0	0	0	0	7	47	213	217	68	1	0	0
	CDH26.7	14	0	0	0	0	0	2	7	4	1	0	0	0
(14) Wind	WSAvg	7.3	9.1	8.0	7.5	7.5	6.7	5.8	5.6	5.7	6.3	7.8	8.7	8.9
(15) Precipitation	PrecAvg	956	81	64	73	84	82	76	87	75	88	85	79	81
	PrecMax	1196	131	135	139	152	168	195	173	133	233	157	151	178
	PrecMin	676	39	35	35	33	32	27	14	21	22	26	19	37
	PrecStd	148	25	24	29	28	35	38	36	27	42	31	30	33
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.6	11.2	16.1	19.9	24.3	26.5	27.5	27.2	26.4	22.5	17.2	13.4
		MCWB	8.4	9.8	10.5	14.7	18.6	19.5	21.6	22.2	21.9	18.1	13.8	11.0
	2%	DB	8.4	8.1	12.1	16.6	21.7	24.7	26.1	26.0	25.0	20.9	15.5	10.6
		MCWB	6.5	6.1	8.1	11.4	17.5	19.5	21.3	21.7	21.8	17.0	12.6	8.5
	5%	DB	6.2	5.6	9.1	14.1	20.0	23.3	25.2	25.2	24.0	19.4	14.1	8.8
		MCWB	4.4	3.6	6.7	10.2	16.1	19.0	21.0	21.2	20.7	16.0	10.9	7.3
	10%	DB	4.2	3.5	6.8	11.8	18.0	22.2	24.5	24.6	23.1	18.0	12.7	7.0
		MCWB	2.6	1.8	4.4	8.0	14.5	18.3	20.6	20.7	19.8	14.8	9.6	5.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.6	10.1	9.5	15.0	19.5	21.8	23.5	23.8	23.2	19.7	15.0	11.0
		MCDB	10.5	11.5	11.1	18.6	23.2	24.3	25.5	26.0	25.3	22.1	16.8	13.2
	2%	WB	6.4	6.7	7.2	12.5	18.2	20.6	22.6	22.8	22.4	18.6	13.2	8.6
		MCDB	7.8	8.2	9.0	16.0	21.1	23.1	24.7	25.0	24.6	20.6	15.4	10.1
	5%	WB	4.3	4.2	5.3	10.5	16.6	19.6	21.9	22.1	21.6	17.3	11.0	7.3
		MCDB	5.6	6.0	7.4	13.4	18.9	21.8	24.1	24.4	23.9	19.7	13.4	8.8
	10%	WB	2.4	2.2	3.4	8.6	14.7	18.6	21.2	21.3	20.7	16.1	9.8	5.1
		MCDB	3.7	3.9	5.2	11.4	17.3	20.8	23.6	23.9	23.0	19.0	12.5	6.7
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	5.3	5.5	6.3	6.2	4.9	3.9	3.7	4.0	4.3	4.5	4.3
		MCDBR	7.2	8.2	9.1	10.6	8.2	5.9	4.3	4.0	4.4	5.2	5.8	5.9
	5% WB	MCWBR	7.4	7.5	7.2	7.8	5.5	4.2	3.3	3.3	3.5	4.3	4.9	6.2
		MCWBR	7.3	8.8	7.9	10.2	8.3	5.3	4.0	3.4	4.2	4.9	5.4	6.2
(40) Clear-Sky Solar Irradiance	taub		0.301	0.345	0.338	0.380	0.406	0.439	0.434	0.422	0.383	0.361	0.315	0.299
		taud	2.417	2.224	2.340	2.311	2.297	2.244	2.282	2.310	2.383	2.419	2.516	2.493
	Ebn at Noon		840	841	905	887	870	838	838	837	846	821	822	815
		Edh at Noon	77	110	111	123	129	136	130	122	106	91	70	66
(44) All-Sky Solar Radiation	RadAvg		1.42	2.10	3.27	4.49	5.56	6.14	6.13	5.40	4.24	2.60	1.60	0.98
	RadStd		0.11	0.24	0.42	0.47	0.54	0.42	0.44	0.29	0.39	0.25	0.24	0.11

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (51 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page